

Python Operators

Operators are particular symbols which operate on some values and produce an output.

The values are known as Operands.

Eg:

1. $4 + 5 = 9$

Here 4 and 5 are Operands and (+) , (=) signs are the operators. They produce the output 9.

Python supports the following operators:

1. Arithmetic Operators.
2. Relational Operators.
3. Assignment Operators.
4. Logical Operators.
5. Membership Operators.
6. Identity Operators.
7. Bitwise Operators.

Arithmetic Operators:

Operators	Description
//	Perform Floor division(gives integer value after division)
+	To perform addition
-	To perform subtraction
*	To perform multiplication
/	To perform division
%	To return remainder after division(Modulus)
**	Perform exponent(raise to power)

eg:

1. >>> 10+20
2. 30
3. >>> 20-10
4. 10
5. >>> 10*2
6. 20
7. >>> 10/2
8. 5
9. >>> 10%3
10. 1
11. >>> 2**3
12. 8
13. >>> 10//3
14. 3
15. >>>

Relational Operators:

Operators	Description
<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than or equal to
==	Equal to
!=	Not equal to
<>	Not equal to(similar to !=)

eg:

1. >>> 10<20
2. True
3. >>> 10>20
4. False
5. >>> 10<=10
6. True
7. >>> 20>=15
8. True
9. >>> 5==6
10. False
11. >>> 5!=6
12. True
13. >>> 10<>2
14. True

15. >>>

Assignment Operators:

Operators	Description
=	Assignment
/=	Divide and Assign
+=	Add and assign
-=	Subtract and Assign
*=	Multiply and assign
%=	Modulus and assign
**=	Exponent and assign
//=	Floor division and assign

eg:

```
1. >>> c=10
2. >>> c
3. 10
4. >>> c+=5
5. >>> c
6. 15
7. >>> c-=5
8. >>> c
9. 10
10.>>> c*=2
11.>>> c
12.20
13.>>> c/=2
14.>>> c
15.10
16.>>> c%=3
17.>>> c
18.1
19.>>> c=5
20.>>> c**=2
21.>>> c
22.25
23.>>> c//=2
24.>>> c
25.12
```

26. >>>

Logical Operators:

Operators	Description
and	Logical AND(When both conditions are true output will be true)
or	Logical OR (If any one condition is true output will be true)
not	Logical NOT(Compliment the condition i.e., reverse)

eg:

1. a=5>4 **and** 3>2
2. **print** a
3. b=5>4 **or** 3<2
4. **print** b
5. c=**not**(5>4)
6. **print** c

Output:

1. >>>
2. True
3. True
4. False
5. >>>

Membership Operators:

Operators	Description
in	Returns true if a variable is in sequence of another variable, else false.
not in	Returns true if a variable is not in sequence of another variable, else false.

eg:

1. a=10
2. b=20
3. list=[10,20,30,40,50];
4. **if** (a **in** list):
5. **print** "a is in given list"
6. **else:**
7. **print** "a is not in given list"

```
8. if(b not in list):
9.     print "b is not given in list"
10. else:
11.     print "b is given in list"
```

Output:

```
1. >>>
2. a is in given list
3. b is given in list
4. >>>
```

Identity Operators:

Operators	Description
is	Returns true if identity of two operands are same, else false
is not	Returns true if identity of two operands are not same, else false.

Example:

```
1. a=20
2. b=20
3. if( a is b):
4.     print ?a,b have same identity?
5. else:
6.     print ?a, b are different?
7. b=10
8. if( a is not b):
9.     print ?a,b have different identity?
10. else:
11.     print ?a,b have same identity?
```

Output:

```
1. >>>
2. a,b have same identity
3. a,b have different identity
4. >>>
```